PARKINOVA

Comprehensive Parkinson's Research Assessment

A. Patient & Assessment Information

Patient Reference	helo
Age	20
Gender	Male
Assessment Date	2026-10-27 09:04:59
Selected Modules	Brain MRI Scans, Archimedean Spiral Analysis, Cognitive Questionnaire
Report Reference ID	55a23e5a-2f4b-4f57-bb78-6ffa061f4309

B. Brain MRI Neuroimaging Analysis

Analysis Status	Completed
Model Prediction	Parkinson's-related pattern detected
Model Architecture	Parkinson CNN (244×244×3)
Input Preprocessing	244×244 Lanczos Resizing, Channel Standardization
Analysis Timestamp	2026-10-27T09:04:59

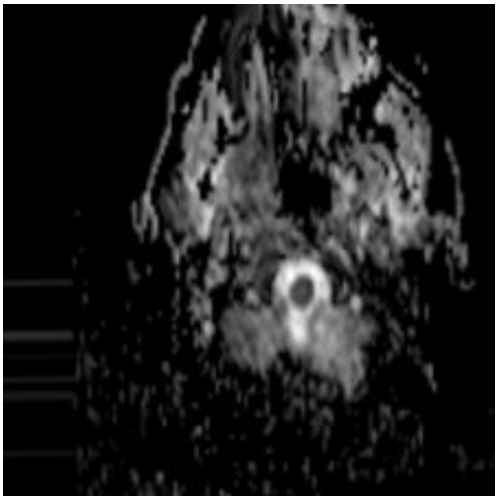
Model Findings & Interpretation:

The CNN deep learning model identified structural and intensity patterns in the MRI scan that correlate with Parkinson's disease neuroimaging features present in the training dataset. This is an automated experimental observation and must be evaluated alongside clinical motor assessments, patient history, and expert radiologist review.

Model Limitations:

The MRI classifier identifies spatial and intensity patterns correlated with the training cohort. An arbitrary MRI scan alone cannot independently confirm or rule out Parkinson's disease. Model confidence is not clinical probability of disease.

Evaluated Brain MRI Scan:



C. Spiral Drawing Analysis

Spiral Analysis Status	Completed (Model Prediction)
Model Prediction	Spiral drawing exhibits regular trajectory without distinctive Parkinsonian tremor pattern
Model Name & Version	Spiral Integration v1.0
Image Resolution	256 × 256 pixels
Canvas Surface	Light background (paper / digital canvas)
Stroke Density	Analyzed

Visual Observations:

• **dimensions:** 256 × 256 pixels
• **aspect_ratio:** 1.0:1
• **stroke_density_pct:** 8.17
• **canvas_style:** Light background (paper / digital canvas)
• **contrast_range:** 255.0
• **file_size_kb:** 98.6
• **Model Status:** Trained Model Active (Parkinson-spiral)
• **Input Resolution:** 244x244 px
• **Raw Model Output:** 1.0000

Analysis Limitations & Recommended Next Steps:

Limitations: Model-based spiral analysis represents pattern classification on isolated drawings. Sensory, visual, or non-Parkinsonian tremors (essential tremor, dystonia) can produce similar visual deviations. Static 2D sketches cannot independently confirm or exclude Parkinson's disease.
Next Steps: Correlate with clinical motor UPDRS examination and neurologist assessment. Consider computerized tablet Archimedean spiral tracking with continuous pen velocity and pressure data.

Uploaded Spiral Drawing:



D. Combined Multi-Modal Assessment Summary

Concordant Multi-Modal Findings: Both the deep learning MRI model and the spiral drawing evaluation observed features correlated with Parkinsonian patterns. In clinical research cohorts, concordant structural and fine motor deviations represent an indication for prompt in-person neurological examination, DaTscan/PET functional imaging, and formal UPDRS rating.

E. Parkinson's Health, Nutrition & Movement Education

Evidence-based supportive lifestyle strategies tailored for individuals with Parkinson's and clinical cohorts:

1. Indian Nutrition & Antioxidants	Focus on neuroprotective, antioxidant-rich foods: cooked spinach (palak), methi, drumstick sambar, ripe papaya, soft ragi kanji/dosas, yellow moong dal khichdi, and soaked almonds. Avoid hard, dry, or splintery textures.
2. Levodopa & Protein Coordination	Dietary large neutral amino acids (LNAAs) compete with levodopa for transport across gut and blood-brain barrier. Take Levodopa 30-60 minutes before or 1-2 hours after protein meals. Do NOT eliminate essential dietary protein.
3. Hydration & Gut Motility	Autonomic constipation affects up to 80% of individuals with Parkinson's. Maintain 6-8 glasses of fluid daily (lukewarm jeera/cumin water, tender coconut water) and adequate soluble fiber (oats, stewed prunes).
4. Safe Physical Activity & Fall Prevention	Daily walking with rhythmic auditory/counting cues (1-2-1-2 cadence), daily calf/chest stretching, supervised balance exercises, and proactive home fall-proofing (removing loose throw rugs, installing night lights).

Regulatory Notice & Academic Disclaimer

This application is an academic AI research prototype. Its output is generated by a machine-learning model and is not a medical diagnosis. The model's performance and reliability may vary depending on image quality, dataset characteristics, and the images used during training. Consult a qualified neurologist for medical evaluation.